Anomaly Localization using Diffusion Models

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Introduction to Diffusion Models

- Inspired by non-equilibrium thermodynamics[1], diffusion models gradually perturb data with noise, in an iterative forward diffusion process.
- The aim is to learn a reverse diffusion process , which removes the addition of noise generating thus realistic samples.
- Due to their impressive generative abilities (better than GANs, VAEs,etc.) diffusion models have been used even in the demanding domain of medical imaging and in applications like anomaly detection.

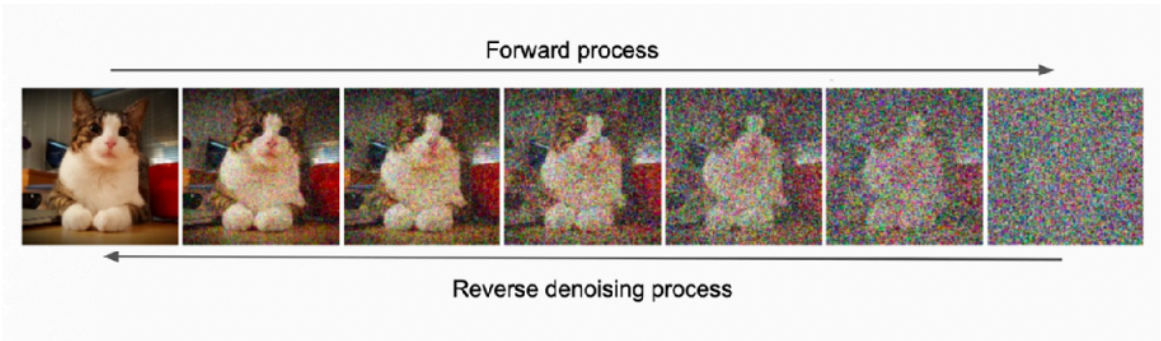


Figure: Diffusion Process

[1] Jascha Narain Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics.
ArXiv, abs/1503.03585, 2015

Essential Concepts of Diffusion Models I

- In the **forward diffusion process** of a classic DDPM[2] the forward process is characterised by a **Markov chain**, which generates a sequence of random variables $x_1, x_2, \ldots, x_T$ with transition kernel $q(x_t|x_{t-1})$.
- We define a variance schedule $\boldsymbol{\beta} = (\beta_1, \beta_2, \ldots, \beta_T)$ such that the noise perturbing the data at each discrete timestep t is an isotropic Gaussian with variance β_t.
- The Markov chain is defined by the transition kernel:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

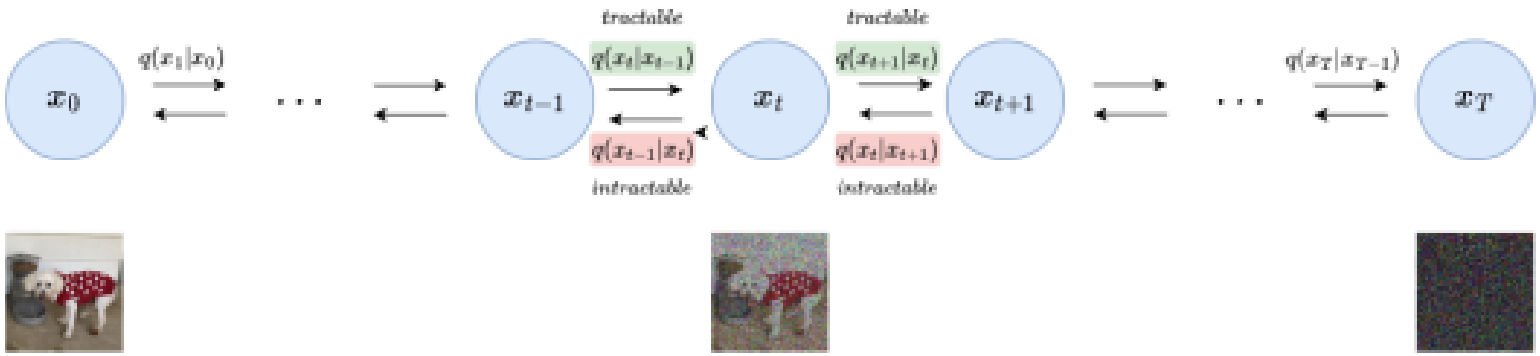
Essential Concepts of Diffusion Models II

- In the **reverse process**, we start from a sample $x_T \sim \mathcal{N}(0, I)$, and generate new samples from $p(x_0)$ by following the reverse steps
- In order to approximate the reverse steps, a neural network is trained that receives a noise input and learns to predict the mean and covariance:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$$

Essential Concepts of Diffusion Models III

- Key to the model's success is the minimization of the Kullback-Leibler divergence between the forward and reverse distributions, ensuring accurate and high-quality generation.



[2] Jonathan Ho, Ajay Jain, and P. Abbeel. Denoising diffusion probabilistic models. *ArXiv*, abs/2006.11239, 2020

Counterfactuals

Diffusion models have been efficiently applied in the process of **pathology localization** by synthesizing **image counterfactuals.**

Image counterfactuals

In image based deep-learning a counterfactual method seeks the segments of the image that when altered would change the classification or the prediction of an AI model to a predefined one.

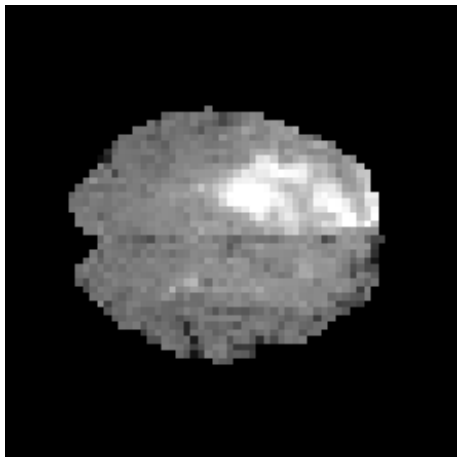


Figure: Original Unhealthy Brain

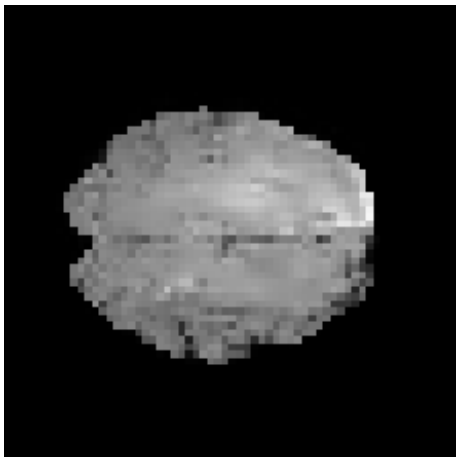


Figure: Healthy Counterfactual

Original Paper Review I

What is healthy? Generative Counterfactual Diffusion for Lesion Localization [3]

- **Goal:** Determine the key features needed to be changed, in order to alter the input image from an unhealthy to a healthy state during the inference phase, while keeping all other elements of the image constant
- **Metric:** Dice Score measures the similarity between the predicted segmentation image with the ground truth segmentation mask on a pixel-by-pixel level. Ranges from 0–1, with a score of 0 indicating no overlap and a score of 1 indication perfect overlap
- **2-step Algorithm:**
  - Encoding is done by iteratively applying diffusion models to obtain a latent space with an unconditional model
  - Decoding is performed by reversing the diffusion process from the latent space to reconstruct an image with healthy condition

Original Paper Review II

- A heatmap of the lesion can be obtained by taking the difference between the original and the reconstructed healthy

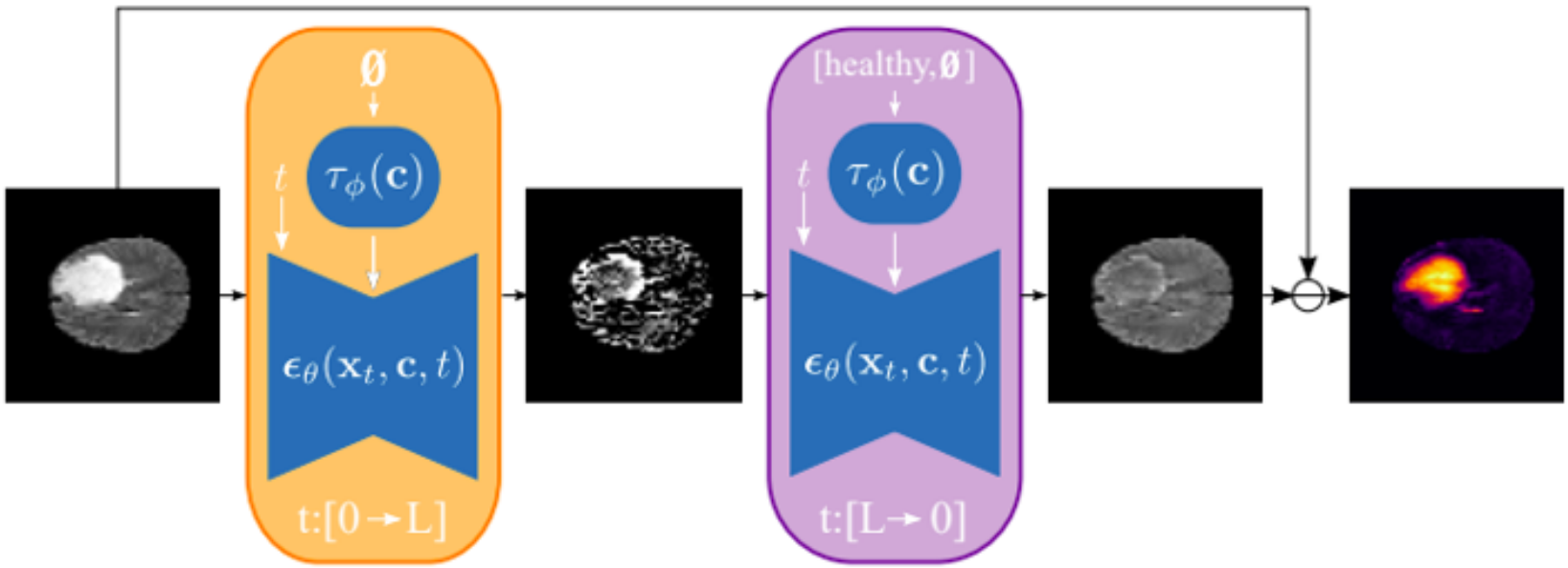


Figure: Counterfactual generation with diffusion models overview

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[3] Pedro Sanchez, Antanas Kascenas, Xiao Liu, Alison Q. O'Neil, and Sotirios A. Tsaftaris. What is healthy? generative counterfactual diffusion for lesion localization.
ArXiv, abs/2207.12268, 2022

Our Research Accomplishments

- Successfully **reproduced** the results of the original paper using **MONAI** (Medical Open Network for AI), an open-source, community-supported framework for deep learning in healthcare imaging
- Calculated the **Dice score**[4] by utilizing both the **standard binary mask** (manual threshold) as well as the **Otsu threshold mask** (separates an image into two classes, foreground, and background, based on the grayscale intensity values of its pixels)
- Experimented with **different hyperparameter values** and determined the impact of each one
  - learning rate
  - training iterations
  - sampling timesteps
- Applied algorithm to **Decathlon Dataset**, targeting the **liver** segmentation task

[4] Tom Eelbode, J. Bertels, Maxim Berman, Dirk Vandermeulen, Frederik Maes, Raf Bisschops, and Matthew B. Blaschko.

Optimization for medical image segmentation: Theory and practice when evaluating with dice score or jaccard index.
IEEE Transactions on Medical Imaging, 39:3679–3690, 2020

Paper Reproduction

Model Architecture

The model used in the paper of Sanchez et al. is a typical **Denoising Diffusion Probabilistic Model** with a U-Net architecture. The parameters of the model are the following :

Parameters	**Value**
Input spatial dimensions	2
Input channels	1
Channels in each U-Net layer	(64, 64, 64)
Attention levels in each U-Net layer	(False, True, True)
Residual blocks in each U-Net layer	1
Channels per head	16
Conditioning	True

Dataset

The data used are from the **Brats 2021 challenge**.This dataset comprises magnetic resonance (MR) imaging from four sequences T1, post-contrast T1-weighted (T1Gd), T2-weighted (T2), and T2 Fluid Attenuated Inversion Recovery (FLAIR) for each patient. Here are the data details of each set in our reproduction:

- **Training Set**: 388 greyscale images of 64 X 64 resolution
- **Validation Set**: 96 greyscale images of 64 X 64 resolution

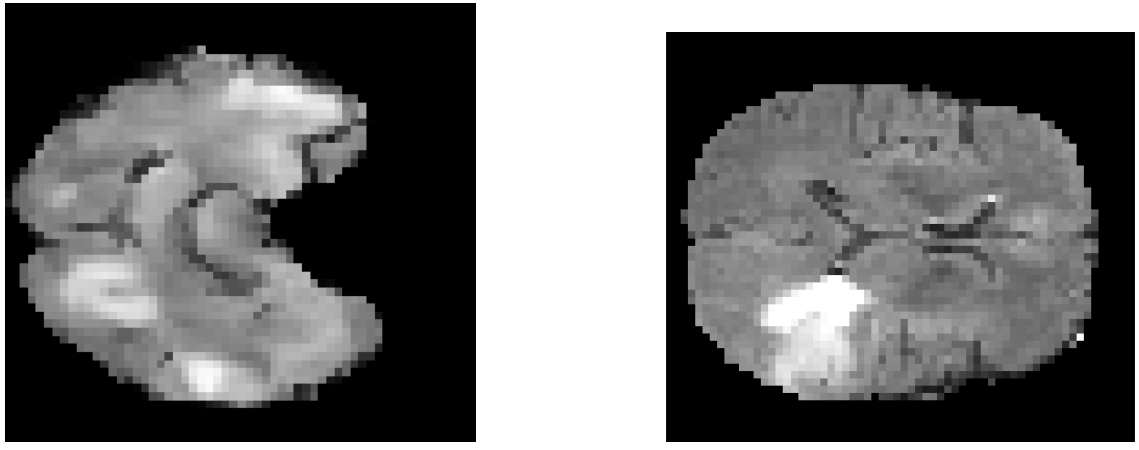


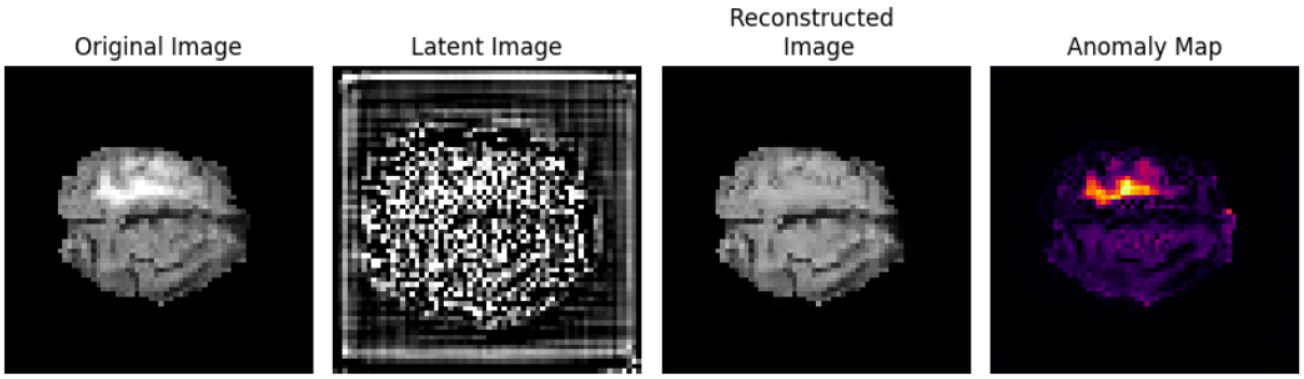
Figure: Images from the Brats 2021 challenge

Original Hyperparameters and Reproduction Results

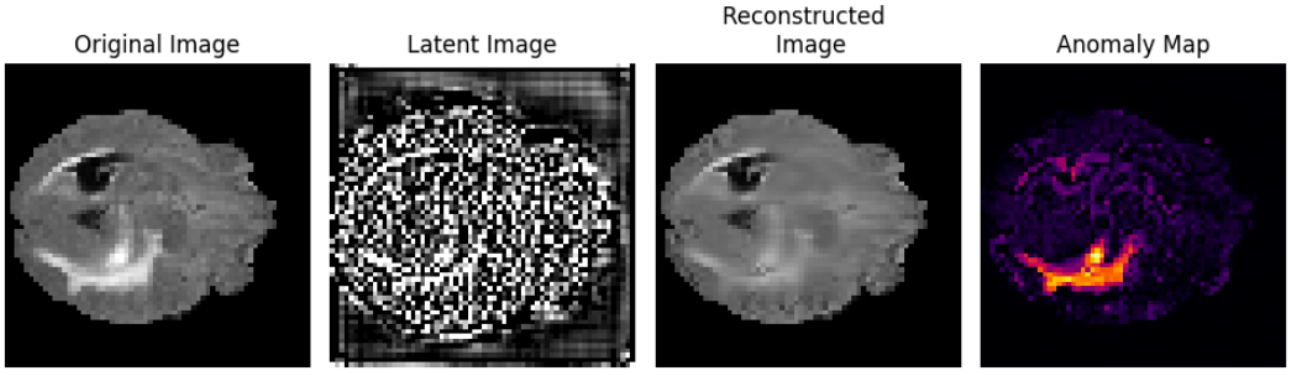
The original paper uses the following hyperparameters:

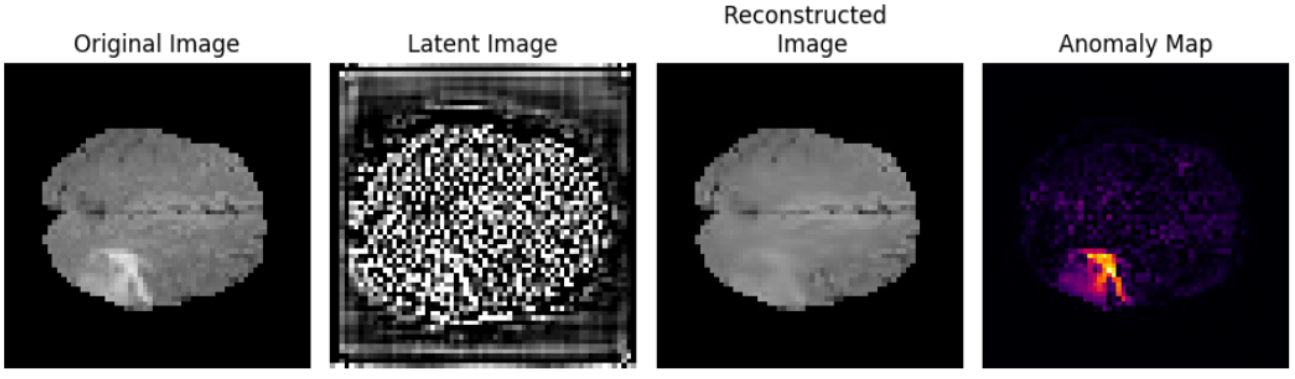
- training_iterations $= 20000$
- learning_rate $= 10^{-5}$
- total_timesteps $= 1000$

Some reproduction results based on the above hyperparameters:

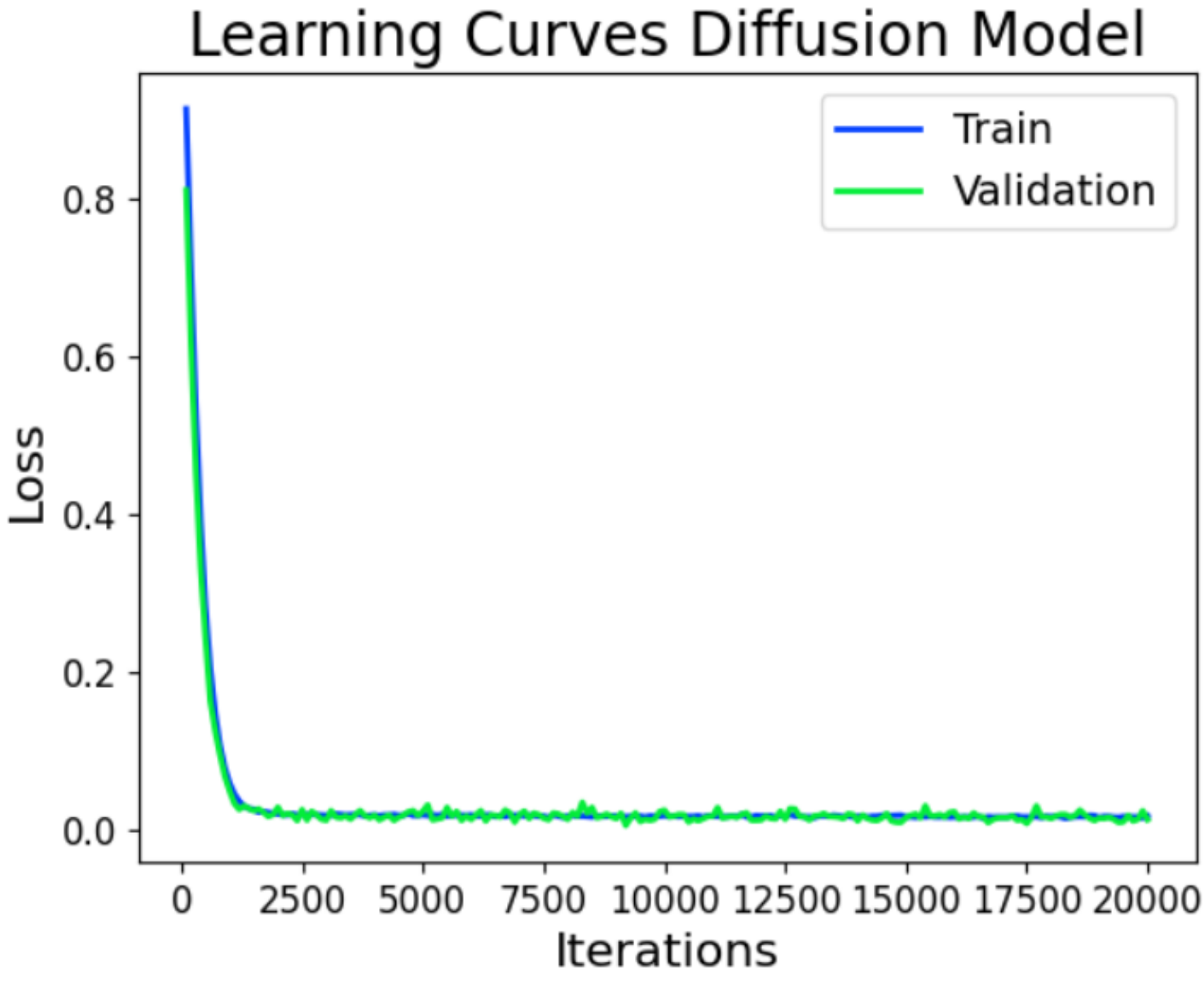


More Reproduction Results



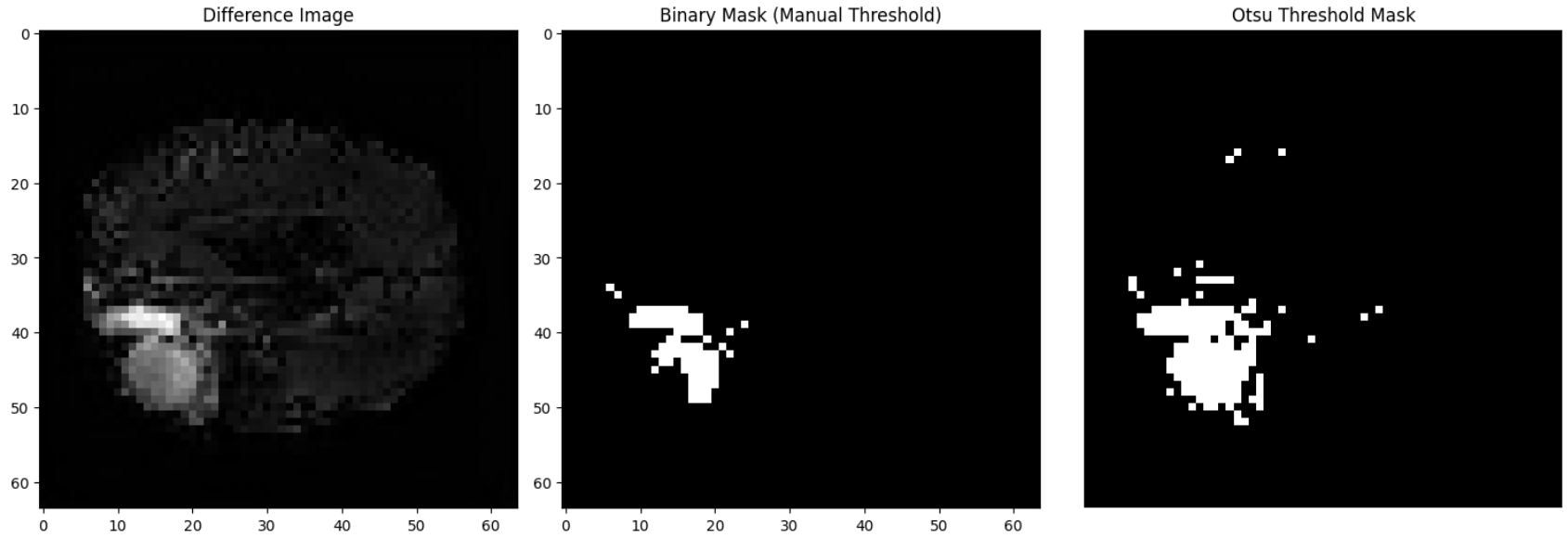


Learning Curve



Dice Score

- **Original Paper:** managed to acquire an average Dice score of **76.2%**
- **Reproduction:** we utilized both **standard binary mask** (manual threshold) as well as the **Otsu threshold mask**
- Standard binary mask: **61.4%**
- Otsu threshold mask: **63.5%**
- We produced lower quality results than the original paper, possibly due to the smaller size of training dataset and the shorter training for our model (limited computational resources available)



Experimentation with Different Hyperparameters

Experimentation with Different Hyperparameters I

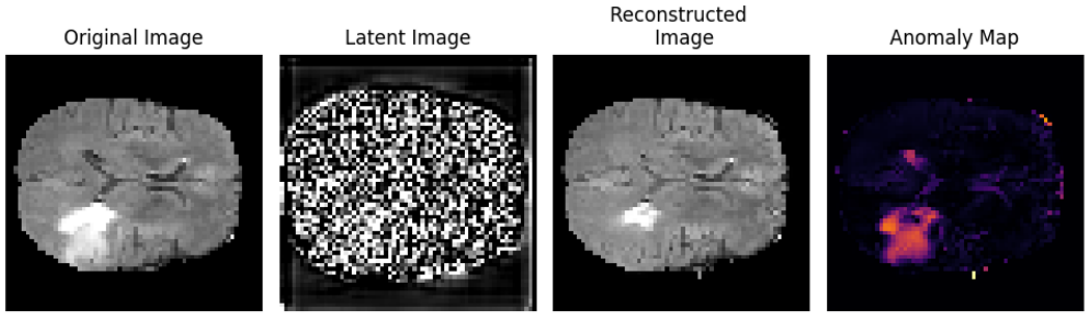


Figure: training_iterations = 5e3, lr=1e-3, total_timesteps = 1000

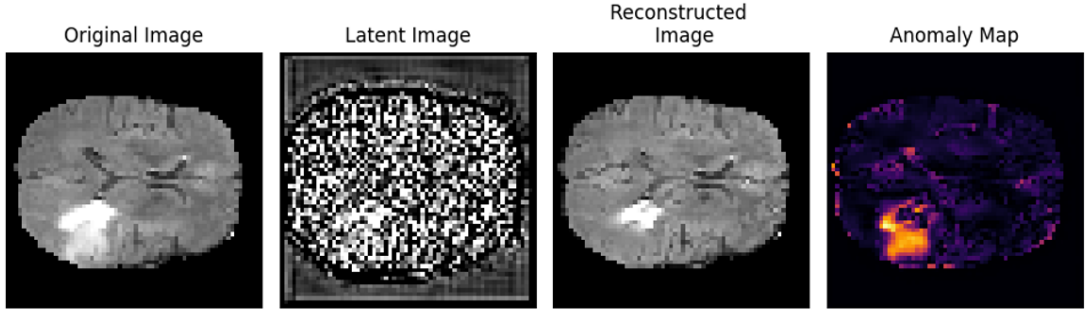


Figure: training_iterations = 5e3, lr=1e-4, total_timesteps = 1000

Experimentation with Different Hyperparameters II

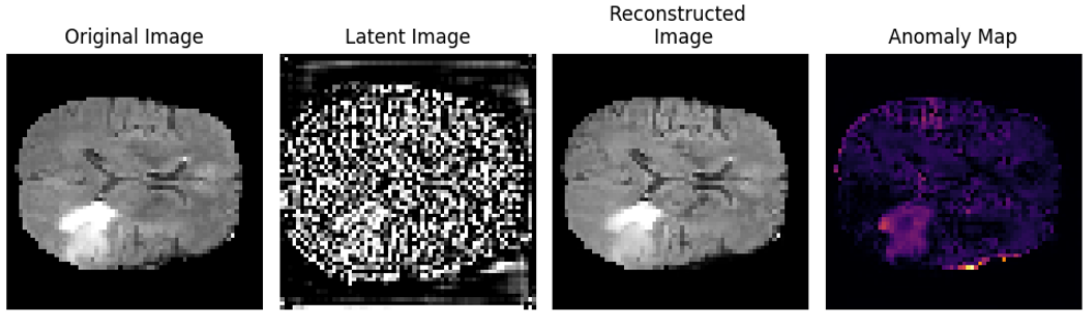


Figure: training_iterations = 5e3, lr=1e-5, total_timesteps = 1000

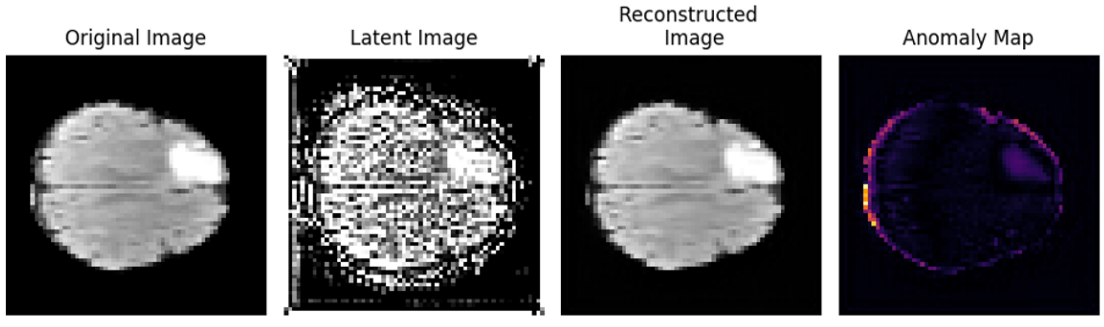


Figure: training_iterations = 2.5e3, lr=1e-5, total_timesteps = 250

Experimentation with Different Hyperparameters III

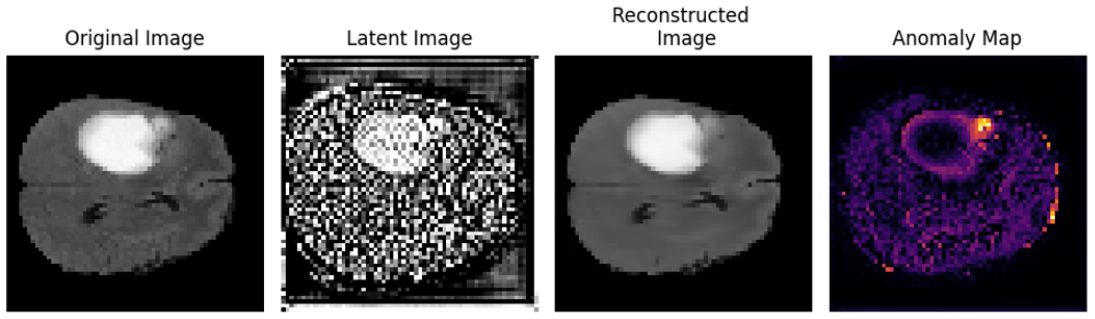


Figure: training_iterations = 1e4, lr=1e-5, total_timesteps = 250

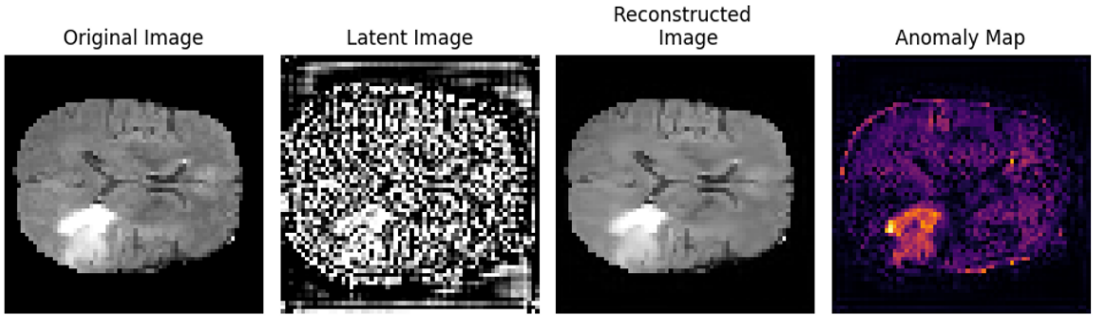


Figure: training_iterations = 5e3, lr=1e-5, total_timesteps = 250

Different Hyperparameters: Conclusions

Based on our experimentation with the hyper-parameters, some patterns emerged regarding each one's relevance in the process:

- Reducing the number of total timesteps in the sampling process results in lower-quality anomaly map images
- Training iterations don't change the model's performance if they exceed 5000
- Learning rates of the order e-4 and e-5 produced the best overall results. Learning rates of the order e-3 and higher produced sub-optimal results

Applying to different dataset

Dataset Description

The new dataset where we aimed to evaluate our model's performance is the one included in the **liver segmentation task** of the **Decathlon Dataset**. It consists of 201 contrast enhanced CT images sourced from patients diagnosed with primary cancers and metastatic liver diseases. Here are the data details of our experiments:

- **Training Set**: 105 grayscale images of 128 X 128
- **Validation Set**: 26 grayscale images of 128 X 128

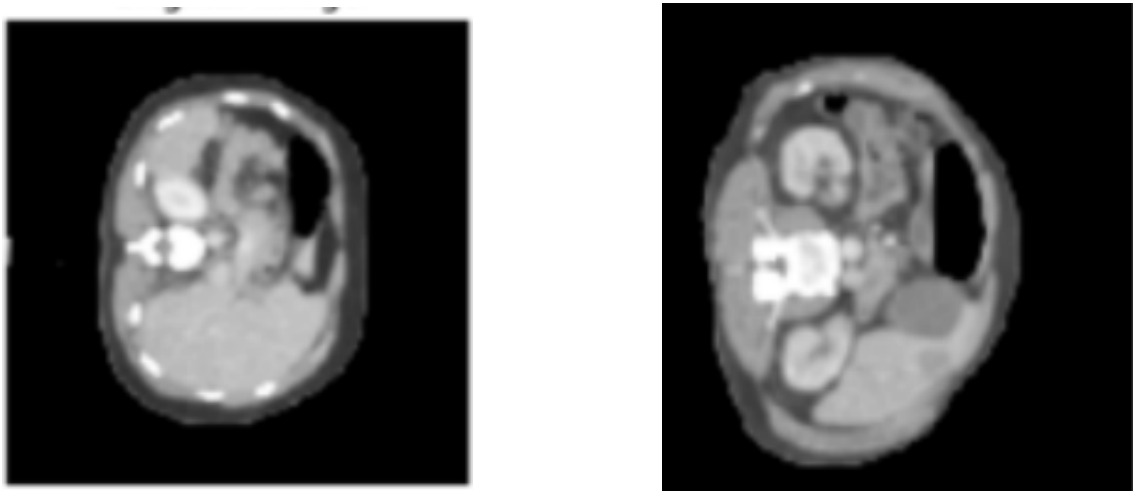


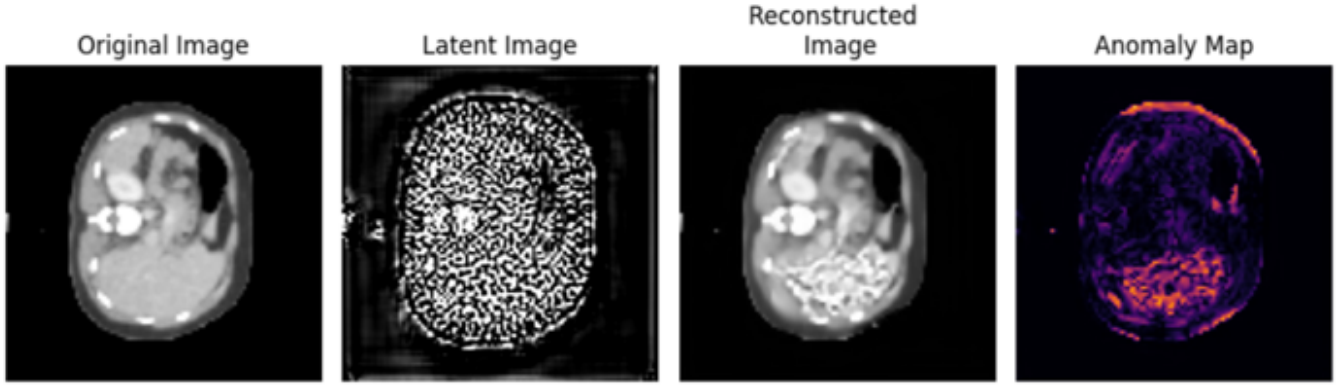
Figure: Images from the liver segmentation task dataset

Hyperparameters used and Results

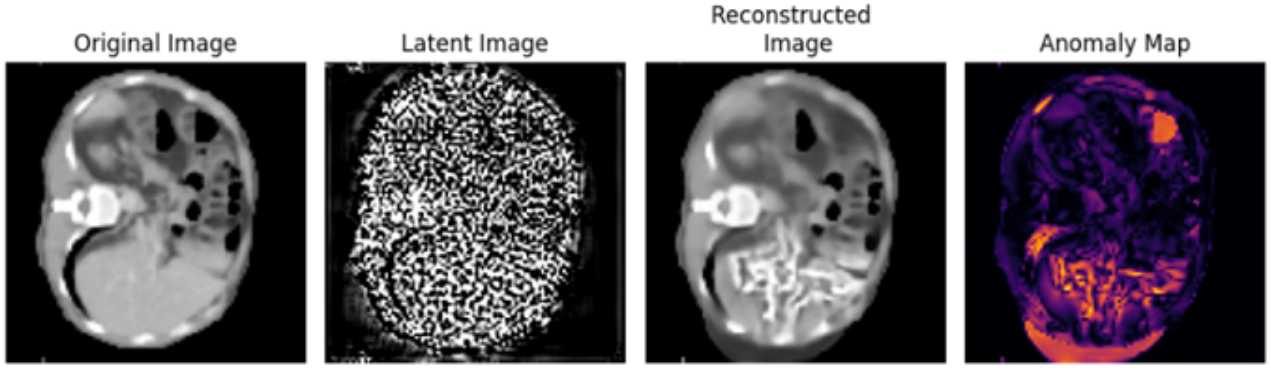
We evaluated our model on the aforementioned dataset using the following hyperparameters:

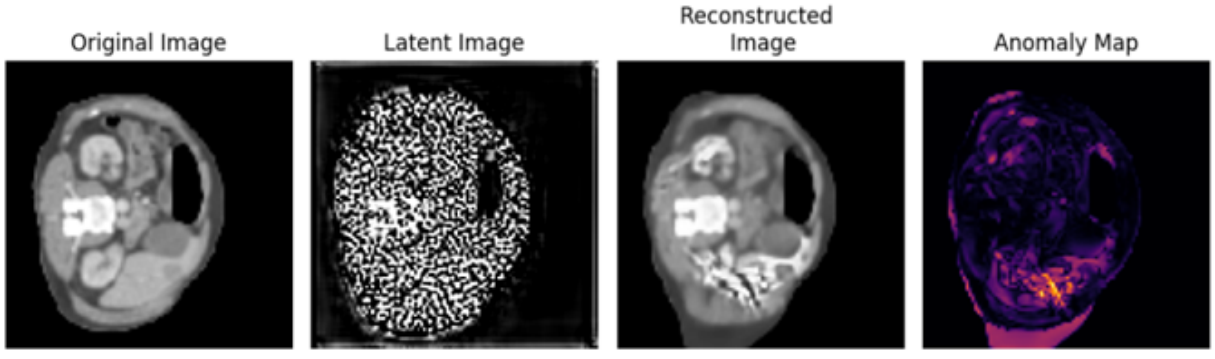
- **training iterations = 10000**
- **learning rate = 1e-5**
- **total timesteps = 1000**

Here are some of our results:



More Results





The average **Dice Score** for the liver anomaly localization task was found to be **51%**

Divergent Results Explained

Observing the above images along with the Dice score, one can clearly see that the results are inferior to the ones we got from the Brats dataset. Here are the possible reasons :

- **Limited Data Availability**: The dataset used for the liver segmentation task has fewer samples than the Brats dataset.
- **Structural Consistency**: The brain has a more consistent structure and shape across various samples compared to the liver
- **Increased Anomaly Diversity**: Liver anomalies demonstrate greater diversity in terms of size, shape and appearance
- **Higher image resolution**: Liver images have relatively higher image resolution

Future Work

- Evaluate the algorithm to various datasets of other body parts with higher resolution images, generating more detailed and precise explanations. This could not be done in this work due to resource limitations
- Experiment with classifier guidance instead of classifier-free guidance as originally proposed
- Evaluate the realism and precision of the generated counterfactuals through a user study of medical experts

Work description

- Pre-existing work:
  - MONAI Implementation
  - Original Paper Implementation from GitHub
- Original Contribution:
  - MONAI Set-up
  - Dice Score with Otsu threshold mask
  - Application to Decathlon Dataset

Members' Contribution

- **Vozinaki**: Experimentation and application to Different Dataset, MONAI framework set-up, Writing Report, Presentation
- **Giadikiaroglou, Litsos**: Experimentation with Different Hyperparameters, Dice Scores, Writing Report, Presentation
- **Karafyllis, Michelakis**: Theoretical Analysis, Paper Reproduction, Writing Report, Presentation

Literature Review - Diffusion Models

Some key papers that have underscored the significance and solidified the dominance of diffusion models in generative modeling, are the following:

- Jascha Narain Sohl-Dickstein, Eric A. Weiss, Niru Maheswaranathan, and Surya Ganguli.

  Deep unsupervised learning using nonequilibrium thermodynamics.
  ArXiv, abs/1503.03585, 2015

- Jonathan Ho, Ajay Jain, and P. Abbeel. Denoising diffusion probabilistic models.
  ArXiv, abs/2006.11239, 2020

- Yang Song, Jascha Narain Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano

  Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations.
  ArXiv, abs/2011.13456, 2020

- Prafulla Dhariwal and Alex Nichol. Diffusion models beat gans on image synthesis.
  ArXiv, abs/2105.05233, 2021

- Jonathan Ho. Classifier-free diffusion guidance.
  ArXiv, abs/2207.12598, 2022

Literature Review - Diffusion Models in medical anomaly detection

Here are some of the most significant papers that have showcased the potential of diffusion models in medical anomaly detection:

- Pedro Sanchez, Antanas Kascenas, Xiao Liu, Alison Q. O'Neil, and Sotirios A. Tsaftaris. What is healthy? generative counterfactual diffusion for lesion localization.
  ArXiv, abs/2207.12268, 2022
- Julia Wolleb, Florentin Bieder, Robin Sandkühler, and Philippe Claude Cattin. Diffusion models for medical anomaly detection.
  In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, 2022
- Alessandro Fontanella, Grant Mair, Joanna M. Wardlaw, Emanuele Trucco, and Amos J. Storkey. Diffusion models for counterfactual generation and anomaly detection in brain images.
  ArXiv, abs/2308.02062, 2023

Thank you for your attention!

Any questions?